

Student Guide — middle / module-13- indic-ecology

IKS Curriculum

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Student Workbook

Student Workbook — Indic Ecology — Vasudhaiva Kuṭumbakam (Middle)

This workbook is yours to write in. Use it for vocab, exercises, and reflection across all 10 days.

Vocabulary tracker

Fill in as we go:

- *Vasudhaiva Kuṭumbakam* — _____
- *yajña* — _____
- *ahiṃsā* — _____
- *devarakāḍu* — _____
- *sthāvara-jaṅgama* — _____

Day-by-day notes

Day 1 — Vasudhaiva Kuṭumbakam

What I learned:

One question I have:

Day 2 — Five Elements as Ecological Frame

What I learned:

One question I have:

Day 3 — Rivers as Sacred

What I learned:

One question I have:

Day 4 — Sacred Groves

What I learned:

One question I have:

Day 5 — Yajña — The Ecological Cycle

What I learned:

One question I have:

Day 6 — Pṛthivī Sūkta (AV 12.1)

What I learned:

One question I have:

Day 7 — Iṣopaniṣad — Non-Greed Ethic

What I learned:

One question I have:

Day 8 — Climate Change — The Crisis

What I learned:

One question I have:

Day 9 — From Ancient to Action

What I learned:

One question I have:

Day 10 — Project Pitch and Assessment

What I learned:

One question I have:

Final reflection

After Day 10, write a paragraph: what is the most important thing you learned in this module?

Day 1 — Vasudhaiva Kuṭumbakam: One Family

Module: Indic Ecology · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will recite the phrase *vasudhaiva kuṭumbakam* with its meaning, name its source (Mahā Upaniṣad 6.71–73), and give one concrete example of “treating something non-human as family” that they have actually seen.

Materials

- Whiteboard with the phrase pre-written in Devanāgarī and IAST
- Printed handout: the half-verse with translation (one per pair)
- Sticky notes (10 per student)
- A large sheet of chart paper labelled “Family” in the centre

Vocabulary introduced today

- *vasudhā* (IAST: *vasudhā*; lit. “the wealth-holder, the earth”) — the Earth as the bearer of all things.
- *eva* (IAST: *eva*) — “indeed, itself.”
- *kuṭumba* (IAST: *kuṭumba*; lit. “household, family”) — a family unit, including extended members.
- *kam* — neuter ending; *kuṭumbakam* = “a family” (the same word with diminutive / endearing tone).
- Combined: *vasudhaiva kuṭumbakam* — “the Earth herself is one family.”

Lesson flow

Warm-up (5 min)

- On the chart paper labelled “Family,” ask each student to write *one* name on a sticky note — someone they consider family — and stick it on the chart.
- After a minute, ask: “How many of these are humans? Anyone include a pet? A grandparent’s tree? A river? A god? A toy?”
- Read 5–6 of the notes aloud without judgement. Most will be human. That’s the data we start with.

Core (20 min)

1. **Write the phrase on the board** in Devanāgarī, then IAST, then word-by-word translation:

वसुधैव कुटुम्बकम् — *vasudhaiva kuṭumbakam* — “the Earth itself is one family.”

2. **The fuller verse.** Read the whole half-verse from the handout:

ayaṁ nijaḥ paro veti gaṇanā laghu-cetasām / udāra-caritānām tu vasudhaiva kuṭumbakam //

“This is mine, that is theirs — such reckoning is for the small-minded. For those whose conduct is large, the whole earth is one family.” — Mahā Upaniṣad 6.71–73

3. **Notice the structure.** It is a *comparison*. The verse is not saying “the Earth IS literally one family” the way “*water IS H₂O*” is a factual claim. It is saying: *if* you live with a big enough sense of belonging, *then* you treat the Earth as a family. The phrase is an ethic, not a biological description.
4. **Where the phrase has lived.** The same idea appears in the *Pañcatantra* prologue and the *Hitopadeśa* as a quoted line; the Mahā Upaniṣad is the older scriptural source. The phrase has been adopted as the motto of the Indian Parliament’s entrance hall, and India used it as the theme of the G20 presidency in 2023. It travels well.
5. **The historical caveat — don’t oversell.** Plenty of pre-modern Indian rulers cleared forests, drained tanks, and treated land as something to extract from. The phrase is an *ideal* the tradition put forward, not a description of how everyone actually behaved. We’ll keep this honest throughout the module.

Activity (15 min) — The “extended family” chart

- Pairs work together. Each pair gets a fresh sticky-note pad.
- Prompt: “Add 5 non-human things to the family chart — but only ones you have personally encountered. Not ‘the Amazon rainforest’ if you haven’t been there. The neem tree by the gate counts. The mango you ate at lunch counts. The pigeon on your roof counts.”
- After 5 min, each pair sticks their notes on the chart. The chart now has ~70–80 sticky notes, mostly non-human.
- **Discussion (5 min):** “Look at this chart. If treating these as ‘family’ is what the verse asks, what would change about how you behave?” (Sample answers: not littering, not killing things for no reason, knowing names, paying attention.)

Wrap-up (5 min)

- One-sentence exit ticket on a slip of paper: “Write one non-human thing you now consider part of your extended family — and one specific behaviour that will change because of that.”
- Collect the slips. Read 3–4 anonymously at the start of Day 2.

Student-facing content

Vasudhaiva Kuṭumbakam — “the Earth itself is one family.”

A line from the *Mahā Upaniṣad* (chapter 6, verses 71–73). It is also quoted in the *Pañcatantra* and the *Hitopadeśa*. It became the motto of India’s Parliament hall and the theme of the country’s G20 presidency in 2023.

The full half-verse:

“This is mine, that is theirs — such reckoning is for the small-minded. For those whose conduct is large, the whole earth is one family.”

The verse is an **ethic**, not a biological claim. It tells you what to do, not what is. The next nine days are about what an ecological ethic of “family” actually looks like in practice — in your river, in your trees, in your habits.

Homework

- Add three more sticky-note items to your “extended family” list at home. They must be things you have *personally* encountered today. Bring the list tomorrow.

Differentiation

- **One tier up:** Compare *vasudhaiva kuṭumbakam* with the modern environmental phrase “Earth is our common home” (used by international NGOs and Pope Francis’s 2015 encyclical *Laudato si’*). Are they saying the same thing? What does each emphasise that the other doesn’t?
- **One tier down:** Focus on memorising *vasudhā* (earth) and *kuṭumba* (family). The whole sentence can come on Day 2.

Teacher notes

- Some students will say “but my cat IS family.” Good — affirm that. The verse extends what they already feel. We are not introducing a new feeling; we’re naming one.
- Some students will say “this is just a religious idea — does it count?” Good — name it. Tell them: the verse is religious *in origin*, but the ethic it carries doesn’t require religious agreement. Many cultures have a version of this idea. The lesson is the ethic, not a conversion.
- Don’t say “ancient Indians invented this and the West later borrowed it.” That’s bad history. Multiple cultures arrived at versions of this ethic independently. Indic thought put it in particularly elegant words.

Citation(s) used in this lesson

- **Mahā Upaniṣad 6.71–73** — *vasudhaiva kuṭumbakam* verse. Full Sanskrit text per the Adyar Library edition.
- See `sources.md` for editions used.

Day 2 — Five Elements Revisited

Module: Indic Ecology · **Band:** Middle · **Time:** 45 min

Learning objective

Students re-state the five elements from Module 2, and re-frame each one as a “family member” of the ecological family — so the *vasudhaiva kuṭumbakam* ethic gets concrete, element by element.

Materials

- Whiteboard
- Each student's Module 2 notebook (if available); otherwise a five-row table on the board
- Five posters or sheets, one per element, labelled in Sanskrit + English
- Sticky notes from Day 1

Vocabulary

(All from Module 2 — review, do not introduce as new.)

- *pr̥thvī* (earth / solid)
- *ap* / *jala* (water / liquid)
- *tejas* / *agni* (fire / heat / light)
- *vāyu* (air / gas / movement)
- *ākāśa* (space)

Add today: - *bhūta-saṃskāra* (IAST: *bhūta-saṃskāra*; lit. “element-care”) — a working term we use for “the care we owe each element.” This is not a classical compound; we use it as a teaching shorthand.

Lesson flow

Warm-up (5 min)

- Read 3–4 anonymous exit slips from yesterday — students' “extended family” examples.
- Quick chorus: “*What were the five elements from Module 2?*” — class names them in order.

Core (20 min)

1. Re-draw the five-element chart on the board:

Element	English	One thing it does for us	One way we pollute it
<i>pr̥thvī</i>	earth / solid	Soil for food, ground to stand on	Plastic waste, chemical fertiliser
<i>ap</i>	water	Drink, wash, irrigate	Sewage, industrial effluent
<i>tejas</i>	fire / heat / light	Cook, warm, illuminate	Burning fossil fuels (CO ₂)
<i>vāyu</i>	air	Breath, wind, weather	Vehicle exhaust, factory smoke
<i>ākāśa</i>	space	Room to live, transmit sound	Noise, light pollution, satellite debris

2. **The reframe.** “*In Module 2 we learned the five elements as categories of matter. Today we look at the same five as members of the family.*” Each element gives us something; we owe each one something back.

3. **Why this is an ecological lens.** Modern environmental science uses categories like “atmosphere,” “hydrosphere,” “lithosphere,” “biosphere,” etc. The five-element framework is a sensory, everyday version of the same idea. It says: *the place you live is made of these five. Notice them. Care for each.*

“Earth, water, fire, air, and sky — these five great elements (*pañca-mahābhūta*) sustain all beings.” — paraphrase from Bhāgavata Purāṇa Canto 3.

(Cited as paraphrase to satisfy Middle-band citation rule; the verse cluster is SB 3.26.40–51 covered in Module 2.)

4. **The “give back” idea.** “What does a family member do for another? They give, they take, and they don’t pollute each other’s spaces. We’ll come back to this on Day 5 when we read the Bhagavad Gītā on the yajña cycle.”

Activity (15 min) — Element-as-family poster

- Group of 4. Each group gets one element-poster.
- They fill in:
 - **One concrete way this element gives to us, near our school.** (E.g. *vāyu*: the breeze through the open windows.)
 - **One concrete way we take from this element, near our school.** (E.g. *vāyu*: the parents’ cars idling outside the gate.)
 - **One concrete way to give back to this element, near our school.** (E.g. *vāyu*: ask parents to switch off engines during pick-up.)
- Posters go up on the wall. They stay up for the rest of the module.

Wrap-up (5 min)

- **Exit ticket.** Each student writes on a slip: “The element I will pay closest attention to this week is _____, because _____.”
- Preview Day 3: “Tomorrow we look at the element that has had the most religious attention in India — water, especially rivers.”

Student-facing content

The five elements as family members.

Element	Gives us	We owe it
<i>pṛthvī</i> (earth)	Food, ground, minerals	Less waste, less concrete, more compost
<i>ap</i> (water)	Drink, growth, cleanness	Less waste, less pollution
<i>tejas</i> (fire / energy)	Warmth, light, cooking	Less fossil fuel, more efficiency
<i>vāyu</i> (air)	Breath, weather	Less burning, fewer engines
<i>ākāśa</i> (space)	Room to live, sound to travel	Less noise, less light at night, less stuff

The five elements are not five things “out there.” They are five members of the family the verse calls *kuṭumba*.

Homework

- Walk for 5 minutes anywhere near your home. Notice ONE specific way one element is being mistreated (the dump, the dust, the noise, the leaking tap). Write 3 sentences. Bring tomorrow.

Differentiation

- **One tier up:** Add a sixth row to the table for “biological diversity” (the *jīva-jantu* — living creatures). Is biodiversity a sixth element? Or is it already covered by *prthvī*? Defend either position.
- **One tier down:** Stick to two elements that feel obvious to you (e.g. water and air). The others will sink in over the week.

Teacher notes

- Some students (correctly) will say “polluting space doesn’t make sense — space is just space.” Push: “*Have you ever been kept awake by a neighbour’s loud TV? Have you been blinded by a glaring street light? That’s polluting ākāśa.*” The five-element frame’s strength is that it forces us to think about subtle pollutions too.
- Don’t let “fire” mean “we should stop using electricity.” That’s not the point. The point is that the *tejas* we use is currently mostly from burning fossil fuels; the family relationship is to use less or use cleaner.
- Some students will ask about climate change here. Hold them off — Day 8 is the proper place. Today is the framework, not the data.

Citation(s) used in this lesson

- Paraphrase of Bhāgavata Purāṇa Canto 3 element-cosmology (SB 3.26.40–51) — sourced from Module 2’s RAG cache, not re-quoted in full here.
- See `sources.md`.

Day 3 — Rivers as Sacred: Protection by Reverence

Module: Indic Ecology · **Band:** Middle · **Time:** 45 min

Learning objective

Students explain how treating a river as sacred functioned (sometimes well, sometimes poorly) as a method of protection in the Indic tradition, name the seven traditionally sacred rivers, and locate the nearest river to their own home.

Materials

- Map of India, paper or projected
- Printed handout: the *sapta-sindhu* invocation (a short stanza traditionally chanted at bathing) with translation
- Two photographs: a clean stretch of a sacred river (upper Yamunā in Yamunotri, or the Gaṅgā at Devprayag) and a polluted stretch (Yamunā at Delhi, Gaṅgā at Kanpur)

- Notebook

Vocabulary

- *sindhu* (IAST: sindhu) — “river, ocean.” Also the name of the Indus river system.
- *sapta-sindhu* (IAST: sapta-sindhu; lit. “seven rivers”) — a traditional list of seven sacred rivers.
- *snāna* (IAST: snāna) — “bath, ritual washing.”
- *tīrtha* (IAST: tīrtha; lit. “ford, crossing”) — a sacred site, often on a river.

Lesson flow

Warm-up (5 min)

- Show the two photographs side by side — same river, different stretches.
- Ask: “*Same religion. Same river. Same name. Why does one look sacred and the other look like a sewer?*”
- Take 2–3 student responses. Don’t resolve yet.

Core (20 min)

1. **The seven rivers.** Write on the board:

Gaṅgā · Yamunā · Sarasvatī · Sindhu · Godāvarī · Narmadā · Kāverī

The traditional invocation (read, don’t memorise):

“gaṅge ca yamune caiva godāvari sarasvati / narmade sindho kāveri jale ’smin sannidhiṁ kuru //”

“O Gaṅgā, Yamunā, Godāvarī, Sarasvatī, Narmadā, Sindhu, Kāverī — be present in this water.”

The verse is chanted while pouring even a glass of water before drinking, in many households. The idea: every drop of water carries the dignity of every great river.

2. **What “sacred” did in practice.** When a river is sacred:

- You don’t urinate or defecate in it (taboo).
- You don’t dump bodies (well — you cremate, then immerse ash, with rules).
- You don’t drain it for one farmer’s field at the cost of the village.
- You build steps (*ghāṭ*) to it; you maintain those steps.

When a river is *not* sacred — or when the sacredness gets only ritual attention and not behavioural attention — you get the second photograph.

3. **Where the system worked and where it broke.** Sacred-status protected upper-stretch rivers and small streams in many villages for centuries. It did *not* protect the urban stretches once cities grew, industry came, and population grew faster than the river could absorb. The Gaṅgā at Haridwar (upper) is still relatively clean; at Kanpur (urban industrial) it is among India’s most polluted stretches.

This is the honest pattern: the tradition gave us a *protective attitude*, not a *technical solution*. Both are needed.

4. **The contemporary state — quick numbers.** (Source: Central Pollution Control Board, government of India, annual river-water-quality reports.) Of India's major river stretches monitored, more than 100 are in the "polluted" category (Class C or worse). Sewage is the primary cause. Industrial effluent is the second.

The remedy is being attempted — Namami Gange (the national clean-up programme) is one example. The tradition can supply the *will* (this river is sacred — we must clean it); the engineers supply the *means* (sewage treatment plants, monitoring, accountability).

Activity (15 min) — Find Your River

- Each student writes:
 1. **Name** of the nearest river / lake / large drain to your home.
 2. **One observation** of its current state (clean? polluted? trickle? full?).
 3. **One specific source of pollution** if you have noticed any (a particular drain, a particular factory, a particular dump).
 4. **Is it considered sacred locally?** (Yes / no / don't know.)
- Pairs share. Two pairs are picked to share with the class.
- The teacher catalogues the rivers / lakes that come up; this list is used on Day 9 for the walk planning.

Wrap-up (5 min)

- **Formative quiz on a slip of paper (5 questions, 3 min)** — see quizzes/formative.md Part A.
- Collect for grading.

Student-facing content

The seven sacred rivers — *Gaṅgā*, *Yamunā*, *Sarasvatī*, *Sindhu*, *Godāvarī*, *Narmadā*, *Kāverī*.

The invocation chanted before bathing or pouring water:

"O *Gaṅgā*, *Yamunā*, *Godāvarī*, *Sarasvatī*, *Narmadā*, *Sindhu*, *Kāverī* — be present in this water."

In Indic thought, sacred = protected. When a river is sacred, you do not pollute it, drain it, or treat it as private property.

But sacred-status alone has not stopped industrial pollution in the urban stretches. The tradition supplies the *attitude of care*; modern science and policy must supply the *engineering and the accountability*. Both are needed.

Homework

- Find out one thing about your nearest river (or water body) that you didn't know yesterday. Examples: the river's name, where it starts, where it joins another river, whether it is monitored. Write 3 sentences.

Differentiation

- **One tier up:** The Sarasvatī in the *sapta-sindhu* list is a “lost river” — the riverbed has been mapped by satellite (Ghaggar-Hakra system) but the river no longer flows. Look up one source on this and bring it tomorrow. What does it mean to call a *dry riverbed* sacred?
- **One tier down:** Memorise just three river names — Gaṅgā, Yamunā, Kāverī (north, north, south). The other four can come over the week.

Teacher notes

- Some students will challenge the “sacred = protected” claim with the Gaṅgā being so polluted. Affirm the challenge. The lesson is precisely that ritual reverence without behavioural follow-through is insufficient. The tradition gave a strong starting point; we have not lived up to it.
- Avoid politicising the Namami Gange programme. Focus on it as one effort, with whatever results it has had.
- Some classes will have students whose families have done *snāna* at a sacred river. Invite them to share if they want; do not single them out. The lesson is enriched by lived experience.

Citation(s) used in this lesson

- *sapta-sindhu* invocation — a traditional prayer chanted across many Hindu households; the exact text is found in standard prayer manuals (e.g. *Nitya Karma Pūjā Prakāśa*). Not from a single canonical scripture; treated here as a traditional liturgical verse.
- Central Pollution Control Board (Government of India) annual reports on polluted river stretches — paraphrased, not cited verbatim.

Day 4 — Sacred Groves: Community Biodiversity Reserves

Module: Indic Ecology · **Band:** Middle · **Time:** 45 min

Learning objective

Students explain what a sacred grove is (community-protected forest fragment) using two regional terms — *devarakāḍu* (Karnataka) and *sarna* (tribal Eastern India) — and identify the ecological function of these groves as biodiversity reserves.

Materials

- A handful of photographs of sacred groves — the *kāvu* of Kerala, the *devarakāḍu* of Coorg, the *sarna* of Jharkhand. (Internet image search; teacher prepares ~6 images.)
- Printed handout: a 1-paragraph extract from Gadgil & Vartak (1976) on the function of sacred groves
- Map of India with sacred-grove regions marked (teacher prepares)

Vocabulary

- *devarakāḍu* (IAST: *devarakāḍu*; lit. “god’s forest,” Kannada) — sacred forest patch dedicated to a village deity, traditionally protected by community rule.
- *sarna* (*sarnā* in some dialects) — sacred grove in the *Adivasi* belt (Jharkhand, West Bengal, Odisha), centred on the *sal* tree and tribal deities.
- *kāvu* — sacred grove in Kerala (Malayalam).
- *orans* — sacred grove / pasture in Rajasthan.
- *biodiversity* — the variety of living species in a particular area.

Lesson flow

Warm-up (5 min)

- Project one photograph of a sacred grove (dense, untouched, surrounded by cleared agricultural land).
- “What do you notice about this patch of forest compared to what’s around it?” Take 2–3 responses.
- “Why might this small patch have survived when the surrounding forest was cut?” Take 2–3 hypotheses. Don’t resolve.

Core (20 min)

1. **Define a sacred grove.** A patch of forest — anywhere from a few trees to several hectares — protected by community rule because of a religious association: a village deity, a goddess, an ancestor, a snake-shrine, a tree itself. The protection takes the form of taboos: no tree-cutting, no twig-collection, no hunting, sometimes no entry except on festival days.
2. **Where they survive.** Sacred groves exist across India:

Region	Local name	Notes
Karnataka (Coorg)	<i>devarakāḍu</i>	~1,200 documented groves; surveyed by Madhav Gadgil’s team in the 1970s–90s.
Kerala	<i>kāvu</i>	Often associated with snake deities (<i>sarpa-kāvu</i>).
Maharashtra (Western Ghats)	<i>devarāī</i>	Surveyed by Gadgil & Vartak in their 1976 paper.
Jharkhand / Odisha / W. Bengal (tribal belt)	<i>sarna</i>	Centred on <i>sal</i> (<i>Shorea robusta</i>); tied to Adivasi worship.
Rajasthan	<i>orans</i>	Often combined with grazing land for

Region	Local name	Notes
Meghalaya (Khasi Hills)	<i>law kyntang</i>	community livestock. Some of the largest and best-protected groves in India.

3. **Why they matter ecologically.** Read the handout extract:

“Sacred groves are relict patches of original vegetation conserved through cultural-religious tradition. They function as in-situ refuges for plant species lost from surrounding areas, including medicinal plants and old-growth trees of high genetic value. In the Western Ghats, sacred groves harbour species not found in nearby agricultural or even reserve-forest landscapes.” — paraphrased from Gadgil, M. & Vartak, V. D. (1976), “The Sacred Groves of Western Ghats in India,” *Economic Botany* 30(2): 152–160.

4. **The mechanism, plainly stated.** When a community treats a patch of forest as sacred, the patch survives. When the community stops, the patch goes the way of any unprotected land — into agriculture or development. Sacred groves are a *social* protection technology, not a biological accident.
5. **Modern parallel.** Modern ecology has a name for this same phenomenon: “community-managed conservation areas” or “biocultural refugia.” The UN’s Convention on Biological Diversity recognises Indigenous and Community Conserved Areas (ICCAs). The Indic *devarakāḍu* and the *sarna* are some of the oldest documented examples of what global conservation policy now formally recommends.

Frame this carefully: *we are not saying “ancient Indians invented modern conservation.”* We are saying multiple cultures, including the Indic tradition, arrived at similar protective practices for similar reasons. The recognition is mutual, not a claim to priority.

Activity (15 min) — Sacred Grove Mapping

(See [activities/activity-01-sacred-grove-mapping.md](#) for the full sheet.)

In groups of 4: - One student looks up your state’s sacred-grove tradition (Coorg / Kerala / Jharkhand / etc.). One paragraph. - One student lists three species you would expect to find in a 200-year-old undisturbed forest patch in your bio-region. - One student draws what a “school sacred grove” could look like — even a 10-tree patch in a campus corner. - One student writes a one-sentence rule the group would adopt to keep their imaginary grove protected.

Groups present in 30 seconds each.

Wrap-up (5 min)

- **Daily prompt.** “Do you know of any patch of land near you — even a small one — that has been protected for unusual reasons?” (A graveyard. A shrine. A temple compound. A lawyer’s office’s overgrown corner.) Each of these may function as an accidental biodiversity refuge.

- Preview Day 5: “*Tomorrow we read the Bhagavad Gītā on why giving back is built into the ecological cycle.*”

Student-facing content

Sacred groves are patches of forest that survived because a community treated them as religious — and that religious status meant rules against cutting and hunting.

Local names you should know:

- *devarakāḍu* — Karnataka
- *sarna* — Adivasi belt (Jharkhand, etc.)
- *kāvu* — Kerala
- *orans* — Rajasthan

In the Western Ghats, ecologists have documented more than 1,000 sacred groves. Some are only a few trees; some are several hectares. Together they preserve plants and animals that survive nowhere else in their region.

The same idea — community-protected conservation areas — is now formally recommended by the UN’s Convention on Biological Diversity (Indigenous and Community Conserved Areas, or ICCAs).

Homework

- Ask one elder in your family or neighbourhood: “*Was there any tree, grove, pond, or piece of land that was ‘not to be touched’ when you were a child? What happened to it?*” Write what they said in 5 sentences. Bring tomorrow.

Differentiation

- **One tier up:** Find one news story from the last 5 years about a sacred grove being threatened (a road, a mine, a township). Summarise in a paragraph. We’ll add it to Day 7’s discussion.
- **One tier down:** Just remember two regional names (*devarakāḍu* and *sarna*) and one fact: sacred groves preserve species the surrounding land has lost.

Teacher notes

- Some students will challenge: “*That’s superstition. Why should we protect a forest because of a goddess?*” — affirm the question. Answer: it doesn’t matter *why* a community protects a forest; what matters is that it gets protected. The Indic tradition used the language of the sacred. Modern conservation uses the language of “ecosystem services.” Both are descriptions of the same protective behaviour.
- The Coorg *devarakāḍu* network is well-documented; the figure of ~1,200 groves is from Gadgil’s published surveys. If you want a source, Gadgil & Subash Chandran (1991) is a more recent count.
- Don’t romanticise sacred groves as universally protected. Many have been encroached, paved over, or turned into commercial temples. The protection is real where it is sustained; it is not magical.

Citation(s) used in this lesson

- **Gadgil, M. & Vartak, V. D.** (1976). “The Sacred Groves of Western Ghats in India.” *Economic Botany* 30(2): 152–160. — Paraphrased in the handout.
- See `sources.md` for the full reference and the Chapple & Tucker (2000) anthology if you need teacher background.

Day 5 — *Yajña*: The Ecological Cycle (Gītā 3.10–14)

Module: Indic Ecology · **Band:** Middle · **Time:** 45 min

Learning objective

Students read Bhagavad Gītā 3.10–14 in translation, identify the four-stage circulation (action → sacrifice → rain → food → beings → action), and articulate why “give back” is *built in*, not optional.

Materials

- Printed handout: BG 3.10–14 in IAST with translation, one per student
- Coloured markers (or four colours of sticky notes) — for diagramming the cycle
- A4 paper, one per pair, for the cycle-diagram

Vocabulary

- *yajña* (IAST: yajña; lit. “offering, sacrifice”) — a reciprocal ritual or action of giving. In an ecological reading, every act that gives something back to the system from which we drew.
- *karma* (IAST: karma; lit. “action, work”) — any deliberate action.
- *parjanya* (IAST: parjanya; lit. “rain, rain-cloud”) — rain treated as a being / process in the cycle.
- *anna* (IAST: anna; lit. “food”) — food, sustenance.
- *bhūta* (IAST: bhūta) — being, living creature. (Not the same use as in *pañcabhūta*, where it means element.)

Lesson flow

Warm-up (5 min)

- Daily prompt on the board: “*Who in nature gave you something today? Did you give anything back?*” Two students share.
- Bridge: “*Today’s text gives you a 3,000-year-old answer to that question — and the answer is: giving back is not extra. It’s the cycle.*”

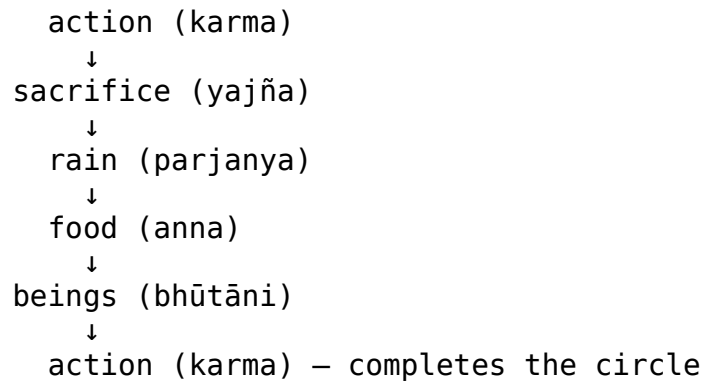
Core (20 min)

1. **Read BG 3.14 first** (the line every student should walk out remembering):

“*annād bhavanti bhūtāni parjanyaḥ anna-sambhavaḥ / yajñād bhavati parjanyaḥ yajñaḥ karma-samudbhavaḥ //*”

“Beings come from food. Food comes from rain. Rain comes from sacrifice. Sacrifice comes from action.” — Bhagavad Gītā 3.14

2. **Diagram the cycle on the board.** Draw four nodes in a circle, with arrows:



3. **Read BG 3.10–13 as context** (handout). Key ideas:

- The act of creating beings included the act of *yajña* — they were created *together*.
- The being who eats food without offering it first “eats theft.”
- Those who eat the leftovers of *yajña* are freed of fault.

4. **The ecological re-reading.** The classical reading is ritual: *yajña* meant a fire offering with specific Vedic formulae. But the structure of the verse is more general:

- Action puts something *out* (work, energy, attention).
- That something has to be *given* (to a god, to a guest, to the field, to the river) — not just consumed.
- The system responds (rain comes; food grows; beings live).
- And the beings act, closing the loop.

In modern ecological language: every act of consumption must be balanced by an act of contribution, or the system runs down. Take more than you give → drought, depletion, climate change, soil exhaustion.

5. **Where this differs from “be sustainable.”** The Gītā does not say *be a little less wasteful*. It says: *whatever you take, give back deliberately, or the cycle breaks*. The verse is structural, not moralistic.

Activity (15 min) — Cycle Diagrams for Real Things

- Pairs pick one everyday consumption and draw the full *yajña* cycle for it:
 - **Option A:** A glass of water from the tap.
 - **Option B:** A plate of rice and dal.
 - **Option C:** A 5-minute scroll on a phone.
- They draw four nodes:

1. *karma* — the action that started this.
 2. *yajña* — what was given (or should be given) to produce this thing.
 3. *parjanya* — the natural process that sustains it (rain, sun, soil, electrical grid).
 4. *anna* — the thing actually consumed.
- For each, they answer: “At which point in the cycle do you, the consumer, give back?”
 - Examples to seed:
 - Water: someone has to maintain the source, the pipe, the treatment plant. Most consumers do nothing.
 - Rice: someone grew it; someone harvested it; the soil sustained it. The consumer’s “yajña” is paying a fair price + not wasting food.
 - Phone-scroll: the algorithm consumed your attention. What did you give back? (Mostly nothing — that’s a real problem.)

Wrap-up (5 min)

- **Mid-module check (formative quiz, 10 min if time allows; otherwise homework).** See quizzes/formative.md Part B.
- One-sentence exit ticket: “One thing I consumed yesterday for which I gave nothing back: _____. One small giving back I will do this week: _____.”

Student-facing content

The *yajña* cycle.

“Beings come from food. Food comes from rain. Rain comes from sacrifice. Sacrifice comes from action.” — Bhagavad Gītā 3.14

The cycle has four stages:

1. **Karma** — deliberate action.
2. **Yajña** — giving something back from that action.
3. **Parjanya** — the natural processes (rain, sun, soil) that respond.
4. **Anna** — the food / sustenance that returns.

...which feeds **bhūtāni** — beings — who do *karma* again. The circle closes.

The Gītā’s claim is that the cycle is *built into how reality works*. Take more than you give, and the cycle breaks. Climate change, soil exhaustion, water tables falling — these are descriptions of broken cycles.

Homework

- Make your own *yajña* diagram for one thing you consume regularly that we did *not* do today (e.g. school uniform, electricity for fan, a packaged snack). Identify the point where you, the consumer, currently give nothing back — and propose one realistic *yajña*.

Differentiation

- **One tier up:** Read BG 3.10–14 alongside one paragraph from a modern ecological-economics source on “externalities” (you can use a Wikipedia summary). Where do the two frameworks describe the same thing in different words? Where do they differ?
- **One tier down:** Memorise just BG 3.14 in translation: “Beings come from food. Food comes from rain. Rain comes from sacrifice. Sacrifice comes from action.” That’s enough for now.

Teacher notes

- Some students will read *yajña* as “religious sacrifice” and shut down. Push: the etymology is just *yaj* — “to offer, to honour, to give.” A meal you cook for a parent is *yajña*. Paying taxes that fund a sewage plant is *yajña*. Sharing food at a festival is *yajña*. The ritual fire-offering is one form, not the only form.
- Avoid teaching this as “the Gītā is about ecology.” It’s not — it’s primarily about *karma-yoga* (the discipline of action). But the cosmological premise embedded in 3.10–14 happens to be ecological in form. We’re reading one strand, not claiming the whole text.
- Watch out for the “and that’s why everything ancient was right” overswing. The verse is a useful frame; it does not, by itself, tell you how to fix sewage.

Citation(s) used in this lesson

- **Bhagavad Gītā 3.14** — quoted in IAST with translation. Text per standard editions; verse cluster 3.10–14 is uncontroversial across recensions.
- Verse context (3.10–13) summarised in the handout, not separately quoted here.
- See `sources.md`.

Day 6 — *Pr̥thivī Sūkta*: Earth as Mother (Atharva Veda 12.1)

Module: Indic Ecology · **Band:** Middle · **Time:** 45 min

Learning objective

Students read selected verses from the *Pr̥thivī Sūkta* (AV 12.1) in translation, identify three concrete images the hymn uses for the Earth, and articulate the difference between *spiritual* and *empirical* claims about the planet.

Materials

- Printed handout: AV 12.1 selections (verses 1, 12, 35, 63) in IAST + translation, one per student
- Notebook
- Optional: a small bowl of soil, a cup of water, a leaf — placed at the front of the room

Vocabulary

- *Pr̥thivī* (IAST: *pr̥thivī*) — Earth, also the goddess Earth. Same word as Module 2’s *pr̥thvī*, here as proper name.
- *sūkta* (IAST: *sūkta*; lit. “well-spoken, hymn”) — a Vedic hymn; one of the units of the R̥g / Atharva collections.
- *mātā* (IAST: *mātā*) — mother.
- *bhūmi* (IAST: *bhūmi*) — earth, land. Used as a synonym for *pr̥thivī*.
- *ṛta* (IAST: *ṛta*) — cosmic order, the way things rightly are. Older companion of *dharma*.

Lesson flow

Warm-up (5 min)

- Show the bowl of soil. Pass around for 1 minute — students smell, touch (not taste).
- “*This is what we are about to call our mother. Hold it. What does that feel like?*”
- Take 2–3 responses. Don’t moralise.

Core (20 min)

1. **Position the text.** The *Pr̥thivī Sūkta* is the longest single hymn in the Atharva Veda — 63 verses — addressed entirely to the Earth. The composition is dated to roughly the late Vedic period (c. 1200–1000 BCE in conventional chronology). It is one of the earliest extensive ecological texts in any tradition.

2. **Read four verses on the handout** (selections — the full 63 are too many for one period):

AV 12.1.12: > “*mātā bhūmiḥ putro ahaṁ pr̥thivyāḥ*” > “Earth is the mother. I am the son of Earth.”

AV 12.1.1 (paraphrase from Griffith / Whitney translation): > “Truth (*satya*), cosmic order (*ṛta*), discipline (*dīkṣā*), spiritual heat (*tapas*), prayer (*brahman*), and sacrifice (*yajña*) — these sustain the Earth. May this Earth, who is mistress of what has been and what shall be, make a wide place for us.”

AV 12.1.35: > “Whatever I dig from you, O Earth, may it grow back quickly. May I not strike your vitals or your heart.”

AV 12.1.63 (concluding verse, paraphrased): > “O Earth, my mother — set me down upon a well-founded place. With heaven for cooperator, O wise one, place me in fortune and in prosperity.”

3. **Notice the imagery.** Ask students to identify, in their notebook:
 - A line that treats Earth as a *person* (mother, who supports, who has vitals).
 - A line that names *what humans owe back* (the dug-up matter should grow back; we shouldn’t strike vitals).
 - A line that lists *abstract things sustained by Earth* (truth, order, discipline, sacrifice).

4. The honest analysis — spiritual claim vs. empirical claim.

Claim	Type	How you'd test it
"Earth is the mother."	Spiritual / ethical	You don't test it. It's a stance.
"Soil takes 100–1000 years to form 1 cm."	Empirical	Soil-science measurement.
"Whatever I dig from Earth, may it grow back."	Ethical (with an empirical implication)	Implies a sustainability check.
"I should not strike Earth's vitals."	Ethical	Implies don't mine recklessly.

The first kind of claim is not better or worse than the second. It does different work. The first shapes your attitude. The second checks whether you actually behaved well.

5. Why this hymn matters now. AV 12.1 is sometimes called the world's first comprehensive ecological hymn. That's a bold claim — but the verse structure does cover:

- Earth as living (mother).
- Reciprocity (give back what you dug).
- Sacred limits (don't strike vitals).
- Sustenance via abstract principles (truth, order).

This is unusual for any tradition's earliest texts. Pre-modern poetry typically extracts; this hymn asks for permission and asks for forgiveness.

Activity (15 min) — Imagery Analysis Worksheet

- In pairs, students fill in a 2-column worksheet:

Verse / image	What modern observation is this in line with?
"Mother who supports us"	Soil supports food chains; the biosphere supports all life.
"May what I dig grow back"	Renewable extraction limits; sustainability.
"Don't strike Earth's vitals"	Don't destroy groundwater aquifers; don't damage the ozone layer; don't push past climate tipping points.
"Earth is the support of truth, order, discipline"	The biosphere is the precondition for any human civilization, including science.

- The point is *not* "the Vedas predicted climate science." The point is that thoughtful pre-modern writers asked questions whose answers we are now measuring. Recognising the overlap honours both sides.

Wrap-up (5 min)

- **Reflection prompt** (slip of paper, 3 sentences): “*Today I read a hymn that calls the Earth my mother. Did this change anything I felt? Be honest — yes, no, or ‘I’m not sure.’*”
- Preview Day 7: “*Tomorrow — what does it mean to enjoy what is given, but not grab? One verse from the Īśa Upaniṣad.*”

Student-facing content

The *Ṕṛthivī Sūkta* — Atharva Veda 12.1.

A 63-verse hymn addressed to the Earth. Some lines you should know:

“Earth is the mother. I am the son of Earth.” (AV 12.1.12)

“Whatever I dig from you, O Earth, may it grow back quickly. May I not strike your vitals or your heart.” (AV 12.1.35)

The hymn is **spiritual** in form — it speaks to the Earth as a person. It is **ecological** in implication — it asks for reciprocity, limits, and forgiveness for what we extract.

Spiritual statements (“Earth is our mother”) and empirical statements (“soil takes 500 years to form 1 cm”) do different work. The first shapes how we *attend*. The second tells us whether we are actually behaving well. Both matter.

Homework

- Read all four verses on your handout one more time. Write **one verse of your own** — 2 lines, addressed to one specific bit of nature near your home (a tree, a tap, a patch of grass). Use the *Ṕṛthivī Sūkta* structure: “*You are X. I am Y. May I not Z.*”

Differentiation

- **One tier up:** Compare AV 12.1.35 (“may what I dig grow back”) with one principle of “regenerative agriculture” (you can summarise from a 5-minute online search). Are these the same claim or different?
- **One tier down:** Focus on one image — “Earth is the mother. I am her child.” That’s enough to anchor the whole lesson.

Teacher notes

- This is the lesson where students most often slide into one of two unhelpful positions: (1) “this is just poetry, not real,” or (2) “this is divine truth and modern science is missing it.” Neither is right. The hymn is *poetry that carries an ethic*. The ethic happens to be in line with what we now know empirically about how ecosystems work.
- The “may I not strike your vitals” verse (AV 12.1.35) is the most operationally useful one. Don’t skip it.
- Some students will ask if the Earth is “literally” alive. Two honest answers exist: (a) no, in the strict biological sense (the planet doesn’t reproduce, metabolise, etc.); (b) yes, in the Gaia-hypothesis sense (the biosphere as a whole self-regulates). Both are respectable positions. The hymn is closer to (b) in spirit but doesn’t depend on either.

- The full 63-verse hymn is available in Whitney’s *Atharva-Veda Samhitā* translation (Harvard Oriental Series 1905) — public domain. Bring more verses if students want them.

Citation(s) used in this lesson

- **Atharva Veda 12.1.12, 12.1.1, 12.1.35, 12.1.63** — quoted / paraphrased. Translations adapted from W. D. Whitney’s *Atharva-Veda Samhitā* (Harvard Oriental Series 1905, public domain) and R. T. H. Griffith’s *Hymns of the Atharvaveda* (1895–96).
- See [sources.md](#) for the edition note and the Chapple & Tucker (2000) anthology for teacher background.

Day 7 — *Iṣopaniṣad* Mantra 1: Use, Don’t Grab

Module: Indic Ecology · **Band:** Middle · **Time:** 45 min

Learning objective

Students recite *Iṣopaniṣad* mantra 1, identify the three injunctions it contains (treat all as enveloped by the divine; enjoy by renouncing; do not covet), and apply the “use, don’t grab” ethic to one specific consumption habit of their own.

Materials

- Printed handout: *Iṣopaniṣad* mantra 1 in IAST + translation, one per student
- Notebook
- Whiteboard

Vocabulary

- *īśa* (IAST: *īśa*; lit. “lord, controller”) — the supreme principle / divinity. The Upaniṣad is named after the first word of mantra 1.
- *īśāvāsyā* (IAST: *īśāvāsyā*) — “to-be-enveloped-by-the-lord,” i.e. permeated by the divine.
- *jagat* (IAST: *jagat*; lit. “what moves”) — the moving / changing world.
- *tyakta* (IAST: *tyakta*; lit. “renounced, given up”) — what is given up, set aside, surrendered.
- *bhujjīthāḥ* (IAST: *bhujjīthāḥ*) — “you may enjoy / partake / use.”
- *grdhaḥ* (IAST: *grdhaḥ*; lit. “grasping, coveting”) — greed, desire to grab.
- *ahiṃsā* (IAST: *ahiṃsā*; lit. “non-harm”) — non-injury to living beings; the first of Patañjali’s five *yamas*. (Bridge from Module 8.)

Lesson flow

Warm-up (5 min)

- On the board: “Name one thing you used today that wasn’t yours.” (Air. Sunlight. The road. The bench. The teacher’s time.)

- Take 4–5 quick answers.
- “Now — did you ‘grab’ it, or ‘use’ it? What’s the difference?” Don’t resolve. The mantra resolves it.

Core (20 min)

1. **Read the mantra together** — students follow along on the handout:

“*īśāvāsyam idaṁ sarvaṁ yat kiñca jagatyāṁ jagat / tena tyaktena bhuñjīthā mā gṛdhaḥ kasya svid dhanam //*”

“All this — whatever moves in this moving world — is enveloped by the Lord. By that renunciation, enjoy. Do not covet anyone’s wealth.” — Īśa Upaniṣad, mantra 1

2. **Three injunctions in one verse.** Number them on the board:

1. **All this is enveloped by the Lord.** Nothing is yours by default. Everything is, in some sense, on loan.
2. **By that renunciation, enjoy.** Use what is given. Take what is needed. Don’t claim ownership beyond what you actually use.
3. **Do not covet anyone’s wealth.** Don’t grab what is not yours — neither another person’s, nor another being’s, nor nature’s.

3. **The phrase *tena tyaktena bhuñjīthāḥ*** is the most quoted half-line. Two readings exist among traditional commentators:

- **Śaṅkara’s reading:** “renounce, then enjoy.” First give up the sense of ownership; then use.
- **Other commentators:** “enjoy what is left over.” First the gods, the ancestors, and the community get their share; then enjoy what remains.

Both readings produce the same ecological ethic: *use, do not grab*.

4. **Why this is ecology.** Modern environmental crises — climate, biodiversity loss, water shortage — are not crises of “use.” They are crises of “grab.” Every human civilisation has used the environment. Most cultures, including the Indic tradition, accepted that use as necessary. But the Indic tradition (in this verse) sets a clear limit: *use is fine, grab is not*.
5. **Link to *ahimsā*.** The first of Patañjali’s *yamas* (eight-limbed yoga, from Module 8) is *ahimsā* — non-harm. Indic ethics extends *ahimsā* beyond humans:
 - Don’t kill what you don’t need to kill.
 - Don’t waste what you took.
 - Don’t take from a being’s last resource.

This is why traditional dietary patterns avoided gratuitous killing; why old laws forbade cutting fruit trees during fruiting; why traditional kitchens kept a portion of cooked rice aside for crows.

These are *ahimsā-shaped* practices, not just rituals.

Activity (15 min) — “What do I grab without asking?”

- Each student takes 5 minutes alone with their notebook.
- Prompt: “List five specific things you took today without asking — and decide, for each, whether you ‘used’ it or ‘grabbed’ it.”
- Examples to seed:
 - Took 5 sheets of printer paper for 2 sheets of work. ← Grab.
 - Drank a glass of water at the school cooler. ← Use.
 - Threw away half a plate of food at lunch. ← Grab.
 - Listened to a teacher’s full explanation. ← Use.
 - Cut a flower off the plant for fun. ← Grab.
- Pairs share two from their list. Pairs help each other classify.
- **Whole-class discussion (5 min):** “Of the grabs we listed — which is the easiest one to stop? Which is the hardest? Why?”

Wrap-up (5 min)

- **Exit ticket** (slip of paper): “My one specific habit-grab from now on will be a use, not a grab. Specifically: I will _____ instead of _____.”
- Preview Day 8: “Tomorrow we look at the biggest grab humanity is collectively making — climate change. Then we put the framework to work.”

Student-facing content

Iṣopaniṣad mantra 1:

“īśāvāsyam idaṁ sarvaṁ yat kiñca jagatyāṁ jagat / tena tyaktena bhuñjīthā mā
grdhaḥ kasya svid dhanam //”

“All this — whatever moves in this moving world — is enveloped by the Lord. By that renunciation, enjoy. Do not covet anyone’s wealth.”

The verse gives three rules:

1. Nothing is yours by default.
2. Use what is given. Don’t claim more.
3. Don’t grab from anyone — human, being, or planet.

The Indic tradition extended *ahimsā* (non-harm) to all of life. Don’t harm what you don’t need to harm. Don’t waste what you took. Don’t take from a being’s last resource.

Homework

- Pick one *grab* from your list today. For 24 hours, replace it with a *use*. Be specific. Bring tomorrow a 3-sentence note on whether it worked.

Differentiation

- **One tier up:** Compare *tena tyaktena bhuñjīthāḥ* with the modern concept of “the commons” (a resource shared by a community, with rules against private grabbing).

Elinor Ostrom won the 2009 Nobel in Economics for showing how communities self-organise to manage commons. What does the Upaniṣad add that economics doesn't?

- **One tier down:** Memorise just the second half of the mantra: “*By renouncing, enjoy. Do not covet anyone’s wealth.*” That’s the practical rule.

Teacher notes

- The Sanskrit *bhuñjīthāḥ* is the optative form of *bhuj* — “to enjoy, eat, partake.” Translators often use “enjoy”; “use” or “partake” works equally well.
- Some students will ask, “But this is asking me to give up things — isn’t that asceticism?” Push back gently: no, the verse says *enjoy*. The renunciation is of *grasping*, not of using. You can enjoy your meal; the renunciation is the *attitude* that says “this is mine and only mine.”
- *grdhaḥ* (the second-person prohibition: “do not grab”) shares a root with “vulture” (*grdhra*). The image is sharp. You can use it.
- Avoid moralising. The exercise is about *noticing* the distinction between use and grab. Most adults haven’t done this exercise. The students will do it more honestly than we will.

Citation(s) used in this lesson

- **Īśa Upaniṣad mantra 1** — quoted in IAST with translation. Text per the standard Mādhyandina recension; identical to Śukla Yajur Veda 40.1.
- See `sources.md` for editions used.

Day 8 — Climate Change: What’s Happening and Why

Module: Indic Ecology · **Band:** Middle · **Time:** 45 min

Learning objective

Students explain — in their own words — what climate change is, name three observable effects, distinguish *climate* from *weather*, and articulate what the Indic frameworks they’ve studied can (and cannot) add to a modern response.

Materials

- Whiteboard
- A simple graph (printed or projected): global average temperature since 1850 (the “hockey stick”). Source: IPCC AR6 Working Group I.
- Optional: a 2-minute video clip from the IPCC’s outreach material or a reliable news source on a recent extreme-weather event
- Notebook

Vocabulary

- *climate* — the long-term average of weather in a region; measured over decades.
- *weather* — what’s happening on a particular day or week.

- *greenhouse gas (GHG)* — gases (CO₂, methane, N₂O, water vapour) that trap heat in the atmosphere.
- *carbon dioxide (CO₂)* — the main long-lived greenhouse gas humans emit.
- *anthropogenic* — caused by human activity.
- *IPCC* — Intergovernmental Panel on Climate Change; the UN body that compiles climate science.
- *SDG* — Sustainable Development Goals (UN, 2015). 17 goals; SDG 13 is “Climate Action.”

Lesson flow

Warm-up (5 min)

- Quick poll, hands up: “*Who has heard the phrase ‘climate change’? Who knows what it means in one sentence?*”
- Take 3 student definitions on the board, without correcting yet.

Core (25 min)

1. **Climate vs. weather — the distinction.**

- **Weather:** “*It rained yesterday.*”
- **Climate:** “*The June monsoon now arrives 7 days later than it did 50 years ago.*”

Weather is local + short. Climate is large + long. Climate change means the *long average* is shifting.

2. **What’s happening (the numbers, simply).** Show the IPCC AR6 graph.

- Global average temperature has risen ~1.1°C above the 1850–1900 baseline. (IPCC AR6, 2023.)
- 2023 and 2024 were the two hottest years on record by a clear margin.
- The IPCC’s “Paris-agreement” target is to keep warming below 1.5°C; we are very close to that line.

3. **Why it’s happening.** Briefly:

- CO₂ in the atmosphere has risen from ~280 parts per million (ppm) in the 1800s to ~420 ppm today.
- That CO₂ comes mainly from burning fossil fuels (coal, oil, gas) for energy and transport, from deforestation, and from cement production.
- More CO₂ → atmosphere traps more heat → average temperature rises.

We are not claiming this is the only cause; we are claiming this is the cause that science has identified as dominant since ~1950. The IPCC reports are the authoritative source.

4. **Three observable effects (pick the ones most local to your students):**

Effect	Where students might already see it
Heatwaves	Indian summer is hotter and longer than 30 years ago

Effect	Where students might already see it
and longer hot seasons	
Monsoon disruption	More floods or droughts; less predictable arrival
Glacier retreat	Himalayan glaciers feed Gaṅgā, Yamunā, Brahmaputra — they are shrinking

5. **Where the Indic frameworks help — and where they don't.**

Help:

- The *yajña* cycle (Day 5) is a direct mental model of “what you take, you must give back.” Climate change is what happens when this is ignored.
- The *iṣopaniṣad* mantra (Day 7) — “*use, don't grab*” — is the ethic of a low-carbon civilisation in five words.
- The *Prthivī Sūkta* (Day 6) — “*may what I dig grow back*” — was always the sustainability rule.
- The *vasudhaiva kuṭumbakam* ethic (Day 1) reframes climate as a *family* problem, not a national one.

Don't help (or don't help by themselves):

- Old frameworks don't tell us what 1.5°C means.
- They don't tell us how to design a solar grid, a heat-resistant rice variety, or a more efficient kiln.
- They don't substitute for measurement, monitoring, policy, and engineering.

Honest summary: the Indic frameworks give us the *attitude* and the *language* of restraint. Modern science gives us the *measurements* and the *technology*. The climate crisis needs both.

6. **What India is doing.** Brief mention:

- India has committed (Paris pledges + COP26) to reach net-zero emissions by 2070.
- National solar capacity has grown ~30× since 2010.
- National Action Plan on Climate Change has 8 missions (solar, water, energy efficiency, sustainable agriculture, Himalayan ecosystem, sustainable habitat, green India, knowledge).

This is not a politics lesson; this is just “what's currently happening.”

Activity (10 min) — Pair brainstorm: pitch the principle

- Each pair picks ONE of the four Indic ideas they've studied:
 1. *vasudhaiva kuṭumbakam* (one family)
 2. *yajña* cycle (give back)
 3. *Prthivī sūkta* (Earth as mother)
 4. *Iṣopaniṣad* mantra (use, don't grab)

- They write 3 sentences: “How does this idea help us think about climate change? What does it NOT solve?”
- This is the precursor to the project pitch on Day 10. Pairs should pick whichever principle they want to anchor their project around.

Wrap-up (5 min)

- **Project plan due tomorrow (Day 9 morning).** Each pair must submit a paragraph stating:
 1. Which Indic principle they will anchor on.
 2. What local observation they will make on Day 9’s walk.
 3. What format (poster / audio / action plan) their pitch will take.
- One-sentence exit ticket: “The hardest thing about taking climate change seriously is _____. ” (Honest answers welcome.)

Student-facing content

Climate change in three sentences.

The Earth’s average temperature has risen about 1.1°C since the 1800s. The cause, since about 1950, is mostly humans burning fossil fuels and cutting forests, which puts CO₂ in the atmosphere and traps heat. The result is hotter summers, disrupted monsoons, retreating glaciers, and rising seas.

What the Indic frameworks add:

- *Yajña* cycle: what you take, you must give back.
- *Iṣopaniṣad*: use, don’t grab.
- *Prthivī Sūkta*: may what I dig grow back.
- *Vasudhaiva kuṭumbakam*: the planet is one family — every nation has to act together.

What modern science and engineering add:

- The measurements (we know exactly how much CO₂ is up there).
- The targets (1.5°C, net-zero, ppm).
- The technologies (solar, wind, electric transport, efficient buildings).

Neither side is complete without the other.

Homework

- **Project plan paragraph due tomorrow morning.** See assessments/project-brief.md.
- Also: pick one news headline from the last week that mentions climate change, weather extremes, or pollution. Bring a one-sentence summary.

Differentiation

- **One tier up:** Read the IPCC AR6 Synthesis Report’s “Summary for Policymakers” — first 5 pages only (it’s accessible online). Bring one fact tomorrow.

- **One tier down:** Focus on one observable effect — the one most visible in your region (heatwaves, floods, monsoon shift). One effect, well understood, is enough.

Teacher notes

- Climate change is a politically loaded topic in some classrooms. Stick to the IPCC numbers; they are the consensus scientific source. Avoid the activist framing on either side.
- Some students or parents may dispute the science. The classroom position is: “the IPCC is what we are teaching. If you disagree, the place to argue with the science is with peer-reviewed responses to the IPCC, not in this classroom.”
- Avoid catastrophising. Students at this age can become genuinely anxious about climate news. Frame it as: *this is real, this is serious, this is also being worked on by millions of people including engineers, farmers, and you.*
- Avoid the opposite — false hope. Don’t say “if we all just plant trees we’ll be fine.” Plant trees AND change energy systems AND change consumption AND change policy. The Indic frame supplies one thread of this; not the whole rope.

Citation(s) used in this lesson

- **IPCC AR6 Synthesis Report (2023)** — for global temperature figures, the 1.5°C target, and CO₂ concentrations.
- **UN Sustainable Development Goals (2015)** — for the SDG 13 reference.
- **National Action Plan on Climate Change** (Government of India, 2008, with 2022 updates) — for the 8 missions framework.

These are all open / government sources; full references in `sources.md`.

Day 9 — From Ancient to Action: The Neighbourhood Walk

Module: Indic Ecology · **Band:** Middle · **Time:** 45 min (20 min walk + 25 min in-class)

Learning objective

Students conduct a 15-minute observation walk on or near the school campus, record one specific ecological observation, and draft a single concrete action they will lead — anchored on one Indic principle.

Materials

- Each pair’s project plan from yesterday
- Notebook + pen
- A printed observation sheet (see `activities/activity-02-eco-walk.md`)
- Permission slip already on file for off-campus walk (only if leaving campus; on-campus is fine without)

- Smartphone with camera if available (one per pair maximum, used only for documentation, not for general phone use)

Vocabulary

(All previous; no new terms today. Today is application.)

Lesson flow

Pre-walk briefing (10 min — in class)

1. **Review the project plan paragraph.** Each pair confirms:
 - Their anchor Indic principle.
 - Their walk objective.
 - Their format (poster / 2-min audio / action plan).
2. **Walk rules.**
 - Pairs stay together. No solo walkers.
 - 15 minutes. The teacher's timer is final.
 - Phones in pocket except for one documentation photo per pair.
 - One specific observation, written down on the sheet.
 - Walk in silence for the first 5 minutes. Talk only after.
3. **What “observation” means here.**
 -  Bad: “*There is pollution.*”
 -  Good: “*At the south gate, between 8:00 and 8:15, twelve cars idled with engines running. The exhaust collected in the corner where the watchman sits.*”

Specificity makes the action possible.

The walk (15 min)

- Pairs leave together with their sheet.
- Teacher walks a perimeter to monitor; no specific guidance during the walk.
- Pairs come back to the classroom at 25 min mark.

Post-walk action drafting (15 min — in class)

1. **Each pair writes — on the sheet — their action plan.** Three lines:
 - **What we observed** (specific, with time and place).
 - **Which Indic principle this connects to** (one of the four).
 - **What action we will lead** (specific, doable in the next 14 days).
2. **Action criteria.** An action is acceptable if:
 - It is specific (names a thing to do, a person to talk to, a place to fix).
 - It can be done within 14 days.
 - The pair can actually do it themselves (no “the government should…”).

3. Examples of acceptable actions:

Observation	Principle	Action
Twelve idling cars at pick-up	<i>Yajña</i> — give back to <i>vāyu</i>	A hand-painted “Engine OFF While Parked” sign at the gate
A leaking tap at the science block	<i>Iṣopaniṣad</i> — don’t grab	Report to admin + remind every day until fixed
A dust-heavy corner with no green	<i>Pr̥thivī Sūkta</i> — Earth as mother	Plant two saplings + water for 2 weeks
A pile of plastic at the canteen	<i>Vasudhaiva kuṭumbakam</i> — one family	Set up one labelled segregation bin + 1 week of monitoring

4. Examples of unacceptable actions:

- “*Stop climate change.*” (Too vague.)
- “*Plant 1 lakh trees.*” (Not doable in 14 days.)
- “*The government should ban diesel.*” (Not the pair’s action.)

Wrap-up (5 min)

- Each pair reads ONE line of their action plan aloud — the action sentence.
- Teacher acknowledges each with a short response (clarify, narrow, encourage).
- Tomorrow: 3-minute pitch + summative quiz.

Student-facing content

The walk has three rules: 1. Stay with your partner. 2. Write down one specific observation — with time and place. 3. Draft one specific action you can actually do.

The action criteria: - Specific (says exactly what you’ll do). - Doable in 14 days. - Done by you, not “by someone.”

The Indic principle anchors your *why*. The observation anchors your *what*. The action is the actual *yajña*.

Homework

- Finalise your pitch:
 - 3 minutes max.
 - Both partners speak.
 - Open with the principle. Show the observation. End with the action.
- Bring your artefact (poster / audio file on a device / printed plan).

Differentiation

- **One tier up:** Add a “measurement” to your action — how will you know whether it worked? (E.g. “*By Day 14, 5 fewer cars idling at pick-up.*”) Measured actions are stronger.
- **One tier down:** Stick to one observation and one action. Skip the measurement. The principle + observation + action is enough.

Teacher notes

- Some pairs will plan an action they can’t actually do. Push back kindly during the post-walk drafting. “Who specifically will you talk to? When?”
- Some pairs will plan an action that is too large to do in 14 days. Cut it. “What’s the smallest version of this you can do this week?” The 14-day frame is non-negotiable; the rubric on Day 10 reflects this.
- The “engine off” sign is the most common student action. Encourage variation — water saving, food waste, biodiversity (a pollinator-friendly corner), noise reduction. Module 13 is about the full five elements, not just air.
- If the school is in a context where an outdoor walk isn’t feasible (rain, unsafe neighbourhood), do a “windows and corridors” walk inside the building. Same rules.

Citation(s) used in this lesson

- None new; today’s anchor is the four principles already cited (Days 1, 5, 6, 7).

Day 10 — Project Pitch and Assessment

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will: present project pitches; summative quiz.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *sthāvara-jaṅgama* (IAST: sthāvara-jaṅgama; lit. “immobile (plants, rocks) and mobile (animals) life”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”

- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Sacred Grove (or Local Ecology) Mapping* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Project Pitch and Assessment.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one paraphrase per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Writing Prompts

Writing Prompts — Indic Ecology — Vasudhaiva Kuṭumbakam (Middle)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 200-word response.

1. In 3–4 sentences, explain Indic Ecology — Vasudhaiva Kuṭumbakam to a friend who hasn’t taken this module.

2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare Indic Ecology — Vasudhaiva Kuṭumbakam with a similar concept from another module or from your science textbook. What's similar? Different?
4. Identify ONE claim in this module that you'd want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about Indic Ecology — Vasudhaiva Kuṭumbakam? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?

How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

Homework

Homework — Indic Ecology — Vasudhaiva Kuṭumbakam (Middle)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — Vasudhaiva Kuṭumbakam** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — Five Elements as Ecological Frame** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — Rivers as Sacred** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — Sacred Groves** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Yajña — The Ecological Cycle** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Pṛthivī Sūkta (AV 12.1)** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Iṣopaniṣad — Non-Greed Ethic** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Climate Change — The Crisis** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — From Ancient to Action** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Project Pitch and Assessment** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.